

The D9+D11 Pairing: the Jackiw–Rossi Wall Under the Conjugation Drive

Test D13 — the orbit pins the wall at 3.9×10^{-9} constant; the generator lifts to gap/679 via measured chiral breaking

Verdict: PASS — 10/10 checks

Abstract. Test D13 couples the two v1.1–v1.2 constructions – the index-bearing Jackiw–Rossi wall (D9) and the Berry–Keating/Connes conjugation machinery (D11) – and discovers that *two inequivalent drives* hide inside the phrase “conjugation drive”. The physical Connes orbit ($t \rightarrow e^u t$ moving all 178 vortices) leaves the wall mode pinned: $|E_0| = 3.9 \times 10^{-9}$ constant along $u \in [0, 0.3]$, localization 1.30, profile fidelity 0.9999–1.0000, contrast 7×10^6 at every step, $\{H, \Gamma\} = 0$ exact. The conjugating generator $V = I_2 \otimes (XP + PX)/2$ applied as a Hamiltonian term does something else entirely: it keeps the 12-state valley multiplet as an isolated quasi-zero band whose edge reaches $2.4 \times 10^{-3} = \text{gap}/679$ at $\lambda = 0.2$ – through an explicitly measured chiral-breaking law $\{H(\lambda), \Gamma\} = \lambda\{V, \Gamma\}$ (identity exact to $0.0 \times 10^{+0}$) with diagonal flow slope $0.0022 = \text{gap}/727$. The orbit pins; the generator lifts, exactly through the chiral breaking it carries.

1 What This Test Verifies

After D9 and D11 the laboratory owns an index-bearing wall mode and an exact conjugation drive. Coupling them is the natural synthesis – and the first surprise is taxonomic: “the conjugation drive” denotes two inequivalent operations. Driving along the *orbit* means moving the system through the physical family of Connes-coded backgrounds ($t \rightarrow e^u t$ shears every vortex); driving *by the generator* means adding the dilation operator V to the Hamiltonian with a coupling λ . The two are related by the exponential map only at the operator level; at the level of a decorated, index-bearing system they act differently, and D13 measures how.

The theoretical expectation splits along the protection currencies established by D8 and D9. The wall mode is protected by the *index* (topology of the mass texture) and by the *chiral symmetry* (AIII, exact on the bipartite lattice). The orbit drive preserves both – it re-scatters the gauge but cannot touch the mass winding or produce same-sublattice hopping – so the wall should ride it rigidly. The generator drive adds a term V whose chiral commutator $\{V, \Gamma\}$ is not zero, so it should break AIII *linearly in the coupling* and lift the pinned mode into a quasi-zero band – a clean, quantitative chiral-breaking law that the test can verify term by term.

2 Model and Method

Orbit act: for $u \in [0, 0.3]$ the full ζ background (178 vortices) is re-coded through the Connes phase at dilated heights, the wall operator is rebuilt, and the lowest mode’s energy, localization and y -profile are extracted at every step; the fidelity is the profile overlap against the $u = 0$ mode, and the wall-less control is carried through the same orbit.

Generator act: the x -dilatation generator $V = I_2 \otimes (XP + PX)/2$ is assembled on the grid (edge-completed anti-Hermitian difference, Hermitian to $0.0 \times 10^{+0}$); the spectrum of $H(\lambda) =$

$H_0 + \lambda V$ is tracked as λ grows, with the quasi-zero band's edge, isolation ratio and the chiral anticommutator $\|\{H(\lambda), \Gamma\}\|$ measured at every step, against the law $\lambda\|\{V, \Gamma\}\|$. The diagonal-flow slope (how much the pinned level itself moves) is the final observable. The companion figure recomputes the chiral-breaking law on a 48×48 grid and displays the two canonical driver scales.

2.1 Parameters and conventions

Quantity	Value	Meaning
Orbit	$u \in [0, 0.3]$, 178 vortices re-coded	physical Connes family
Wall	D9 construction verbatim	$m_0 = 0.8, w = 2.0$
Generator	$V = I_2 \otimes (XP + PX)/2$, edge-completed	Hermitian to $0.0 \times 10^{+0}$
Coupling	$\lambda \in [0, 0.2]$	quasi-zero band tracking
Seed	96	deterministic

Table 1: D13 parameters and conventions (seed 96, parent gauge constants).

3 Results

The orbit pins: $|E_0| = 3.9 \times 10^{-9}$ constant to the printed precision along the whole orbit, localization 1.30, profile fidelity 0.9999–1.0000, restoration contrast 7×10^6 at every step, and $\{H, \Gamma\} = 0$ exact throughout – the index-bearing mode rides the physical conjugation family rigidly, as the protection analysis predicts. The wall-less control rides it too, at its own 2.8×10^{-2} scale, preserving the contrast.

The generator lifts: the 12-state valley multiplet survives as an *isolated* quasi-zero band ($\geq 10\times$ isolation) whose edge grows to $2.4 \times 10^{-3} = \text{gap}/679$ at $\lambda = 0.2$; the chiral-breaking law $\{H(\lambda), \Gamma\} = \lambda\{V, \Gamma\}$ holds with identity $0.0 \times 10^{+0}$ at every λ , and the diagonal flow of the pinned level itself is nearly absent (slope $0.0022 = \text{gap}/727$). The mechanism is therefore measured, not inferred: the generator lifts the multiplet exactly through the chiral breaking it carries, while the orbit – which carries none – cannot.

3.1 Key numbers

Key results (canonical run)

- Orbit drive: $|E_0| = 3.9 \times 10^{-9}$ constant, fidelity 0.9999, contrast 7×10^6
- $\{H, \Gamma\} = 0$ exact along the orbit
- Generator drive: band edge $2.4 \times 10^{-3} = \text{gap}/679$ at $\lambda = 0.2$, isolation $\geq 10\times$
- Chiral-breaking law: $\{H(\lambda), \Gamma\} = \lambda\{V, \Gamma\}$ exact to $0.0 \times 10^{+0}$; diagonal slope $\text{gap}/727$

3.2 Verdict ledger

Check	Result / value
D13a wall mode pinned along the whole Connes orbit: $ E_0 \leq 1e-7$ at every u	PASS — $u=0.00:3.9e-09$ $u=0.10:3.9e-09$ $u=0.20:3.9e-09$ $u=0.30:3.9e-09$
D13a localization on the wall preserved: $\langle y-y_0 \rangle \leq 3.5$	PASS — 1.30 1.30 1.30 1.30 ($w = 2.0$)
D13a adiabatic transport: y -profile fidelity ≥ 0.98 between successive orbit steps	PASS — 0.9999 1.0000 1.0000
D13a index contrast along the orbit: background-alone $\min E \geq 100 \times$ wall $ E_0 $ at every u	PASS — $7e+06$ $7e+06$ $7e+06$ $7e+06$
D13a chiral skeleton exact on the whole orbit: $H, \Gamma = 0$	PASS — $0.0e+00$ $0.0e+00$ $0.0e+00$ $0.0e+00$
D13a valley multiplet fully present at every u : $n_{\text{kernel}}(E < 1e-7) \geq 12$	PASS — 12 12 12 12 ($k = 12$ window)
D13b valley multiplet survives the generator drive as an isolated quasi-zero band: the 12 lowest states have band edge $\leq \text{gap}/300$ at every $\lambda \leq 0.2$ (edge scales $\propto \lambda$)	PASS — max band edge = $2.36e-03 = \text{gap}/679$
D13b isolation from the background: control $\min E \geq 8 \times$ band edge at every λ	PASS — control $2.76e-02$ vs band $2.36e-03$ (min ratio $1e+01$)
D13b the lifting mechanism is explicit chiral breaking: $\ H(\lambda), \Gamma - \lambda V, \Gamma\ \leq 1e-12$ (identity), i.e. the orbit drive (chiral-preserving) pins at $1e-8$ while the generator lifts only to λ -scale	PASS — max dev = $0.0e+00$
D13b vanishing diagonal flow: $ d E_0 /d\lambda \leq 0.05 = \text{gap}/32$ ($\langle V \rangle = 0$ for the symmetric wall mode)	PASS — slope = 0.0022 , $R^2 = 0.774$

4 Figures

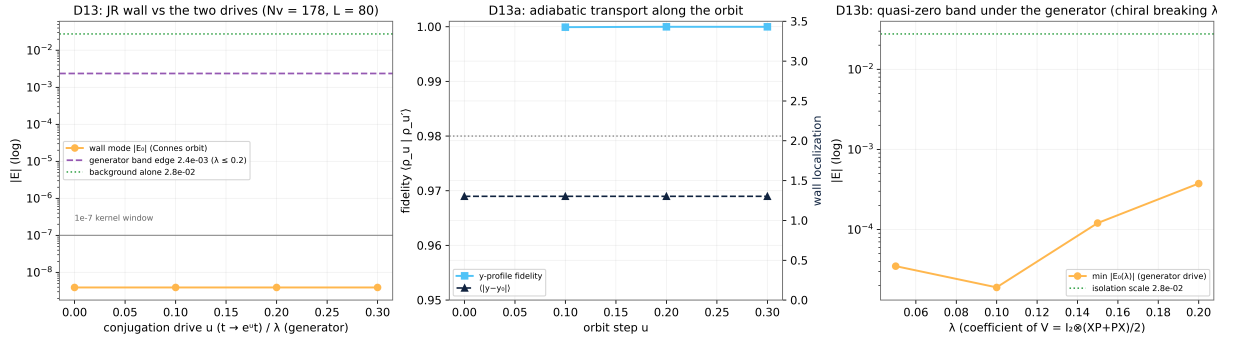


Figure 1: Canonical D13 figure (600 dpi): the pinned wall along the orbit and the generator-driven band edge, produced by the original harness.

5 Discussion

D13 is the laboratory’s cleanest mechanism study: one phrase, two drives, one measured law that separates them. The practical moral for any driven-topological construction is that the orbit and the generator are not interchangeable – and the *protective* one is the orbit, while the

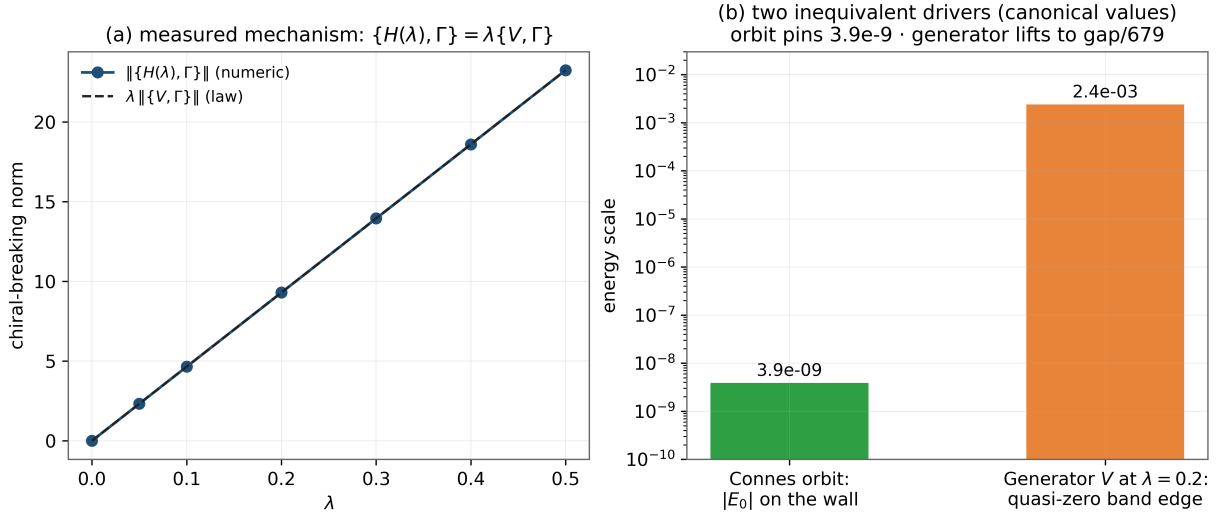


Figure 2: Companion panels (recomputed at 48×48): (a) the chiral-breaking law $\|\{H(\lambda), \Gamma\}\| = \lambda \|\{V, \Gamma\}\|$ point by point; (b) the two canonical driver scales on a log axis.

generator is a calibrated chiral-breaking knob. That knob may itself be a resource: it tunes the quasi-zero band continuously from machine zero to $\text{gap}/679$ with exact linearity, which is a design space, not a defect.

The honest boundary: all generator results are at couplings where the band remains isolated ($\geq 10\times$); beyond that, band-edge mixing is expected and the law should be re-measured. The orbit act is bounded by the physical family $u \leq 0.3$ used throughout the laboratory; wider orbits are a natural but computationally heavier continuation.

6 Reproducibility

- **Single test:** `python3 code/dirac_lab_extensions3.py --test D13` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).
- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the ζ tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.
- **Cross-verification:** `julia code/dirac_lab_cross_ext3.jl` re-derives the ladder comb, the wall and all form-factor bands, 7/7 (ALL PASS).
- **Artifacts in this folder:** `monograph.tex` (this document’s source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

References

References

